

	Continuous variant	Unseen allele	Allele-specific output	Training completed	Allele score defined
<i>Variant-conditionable</i>					
scGen	✓	✓	✓	✓	✓
scVIDR	✓	✓	✓	✓	✓
Biolord	✓	✓	✓	✓	✓
CellFlow	✓	✓	✓	✓	✓
PerturbNet	✓	✓	✓	✓	⚠
<i>Allele-blind by construction</i>					
scGPT	✗	✗	✗	●	✗
GEARS	✗	✗	✗	●	✗
STATE	✗	✗	✗	✓	✗
<i>Did not converge</i>					
variant-CPA	✓	✓	●	✗	✗
✓ available ⚠ caveat ✗ unavailable ● not applicable Gene-keyed coverage 4/470 = 0.85% of benchmark variants representable Status reflects the evaluated implementation					